

Jay Patel

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AI/ML & Data Science | Open Source Contributions - Pandas

EDUCATION

BSc (Hons.) Computer Science

Expected: May 2027

York University, Toronto, Ontario

Advanced Math(Calculus, 3D Geometry, Probability, Statistics), Programming(Python, SQL, Java, C), Data Structures and Algorithms, Artificial Intelligence, Relational databases, Risk Management

EXPERIENCE

Research — AI and Machine Learning, York University

Sep.'25-Dec.'25

- Designed a Reinforcement Learning system using PyTorch, Sentence Transformers, and FastAPI.
- Applied NLP and transformer based reasoning to guide RL rewards, improving code generation accuracy by 15% on 400+ competitive programming problems.
- Analyzed 300+ GitHub logs in Microsoft Excel for faculty research, identifying software bugs.
- Preparing academic submission at ACM International Conference 2026, Montreal.

PROJECTS

Open-Source Contributor — Pandas (10 Million+ global users)

- Bug fix and added test cases for *ExtensionArray* operations; 15+ commits with multiple code reviews. (PR [merged](#))
- Improved CSV parsing with *on_bad_lines*; contributed to iterative development, unit testing and debugging.

Demand Forecasting and Anomaly Detection Dashboard [GitHub](#)

May 2025

- Processed 40M+ NYC taxi trips, cleaned 1.2M+ outliers, and added new features using PySpark, Dask, and Pandas.
- Trained predictive models - XGBoost (13% MAPE) for demand and LSTM (0.055 MSLE) for detecting anomalies.
- Built a Power BI dashboard from model outputs stored in MySQL for easy visualization and analysis.

Data Engineering, Power BI, SQL, Python, XGBoost, LSTM, PySpark, Dask, Pandas, Data Mining, Tensorflow

Quantitative Trading Algorithm with LSTM [GitHub](#)

July 2025

- Built an ensemble trading model with LSTM, Bollinger Bands, and RSI, executing trades when 2+ indicators aligned.
- Achieved +54.6% cumulative return and 6.5% CAGR over 5 years across 50+ stocks.
- Outperformed the S&P 500 by 17%, achieved a 57% return in the best year, and preserved capital during COVID-19 in volatile market conditions.

Time Series Forecasting, Feature Engineering, Statistical Analysis, LSTM, Pandas, NumPy, scikit-learn, TensorFlow, Financial Analysis, Data Visualization (seaborn, quantstats), Data Modeling

Azure Deployed Emotion Detection with Docker [GitHub](#)

March 2025

- Trained a ResNet-based CNN with L2 regularization and batch norm, achieving 75% accuracy on FER2013 (Kaggle).
- Integrated MLflow to monitor model performance and data quality.
- Containerized with Docker and deployed on Azure, featuring a Streamlit + WebRTC app.

Computer Vision, MLflow, Docker, Azure, Python, TensorFlow (Keras API), OpenCV, CNNs, Data Collection

AI RAG Assistant Using LangChain + LLMs [GitHub](#)

September 2025

- Built a context-aware chatbot using LangChain and Watsonx LLM to provide answers from documents.
- Created a document data pipeline with RAG that splits PDFs into 100-token chunks.

IBM Watsonx LLM, LangChain, Retrieval-Augmented Generation (RAG), Semantic Search, Prompt Engineering

SKILLS

Machine Learning Algorithms: Regression, Classification, Clustering, CNNs, LSTM, Reinforcement Learning

LLM & NLP Frameworks: IBM Watsonx, LangChain, LangGraph, Hugging Face, RAG, Open AI API

Libraries: scikit-learn, TensorFlow, Keras, PyTorch, NumPy, Pandas, PySpark, XGBoost, JAX, Matplotlib, Plotly

Languages: Python, SQL, R, Scala, Java

Cloud & DevOps: Docker, Kubernetes, Azure, GCP, CI/CD, FastAPI, Linux, Unix, MLflow, Git, Github

Other tools: Databricks, Apache Spark, Airflow, Tableau, Excel, PowerPoint

CERTIFICATIONS

IBM AI Engineering Professional Certificate

Built and deployed Generative AI and Deep Learning models with end-to-end ML workflows, including data analytics, preprocessing, feature engineering, model training, evaluation and cloud deployment workflows.